

I. Dataset Overview

The Titanic dataset contains information about passengers on the ship, including basic details like their name, sex, and age. It also shows their status on the ship through passenger class and fare. The dataset includes information about family members traveling with them, such as the number of siblings or spouses and parents or children. It also has records of ticket details and the port where each passenger boarded the ship. These details can be used to study patterns that may have affected passenger survival during the tragedy.

II. Univariate Analysis

This section focuses on understanding the basic characteristics of the passengers without considering survival outcomes yet. This includes an overview of how many passengers survived versus those who did not and how passengers are spread across the different ticket classes. These visualizations provide a general picture of the passenger population and help set the foundation for analysis in the next sections.

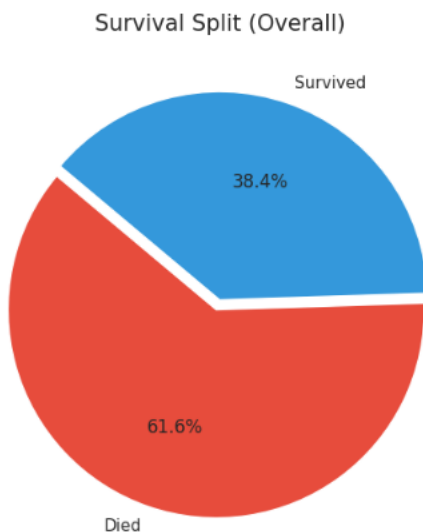


Figure 1.1

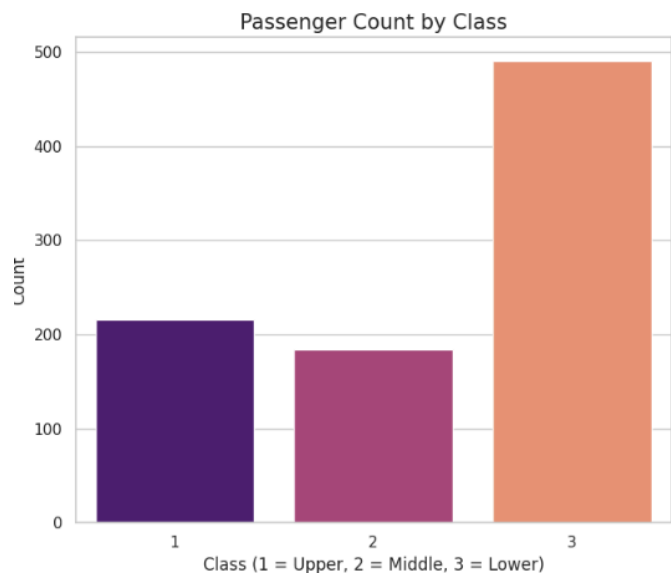


Figure 1.2

Figure 1.1 shows the percentage split between passengers who survived and those who did not. Out of the 891 passengers in the dataset, only 38.4% (342 people) survived, while 61.6% (549 people) died. **Figure 1.2** shows the distribution of passenger classes. The graph indicates that the majority of passengers in the dataset belong to the lower class.

III. Bivariate Analysis

This section explores how survival is influenced by other passenger characteristics. Here, we compare survival rates across genders, age groups, and passenger classes. These comparisons help identify patterns, such as whether women and children were more likely to survive, or whether passengers in higher classes had better chances.

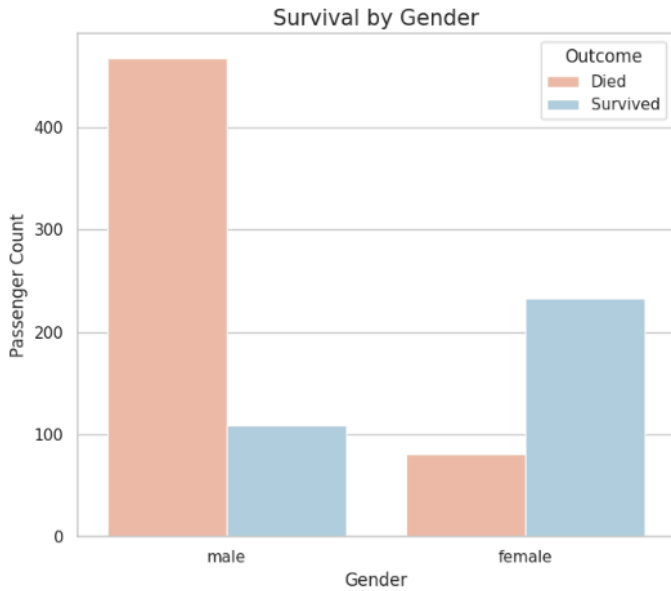


Figure 2.1

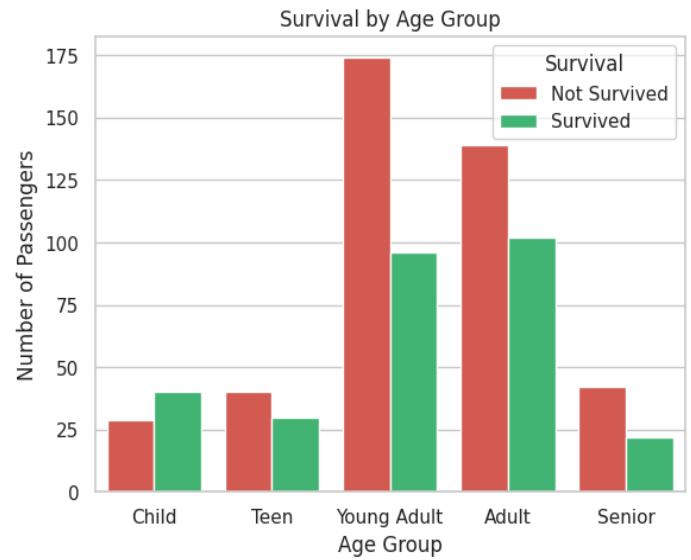


Figure 2.2

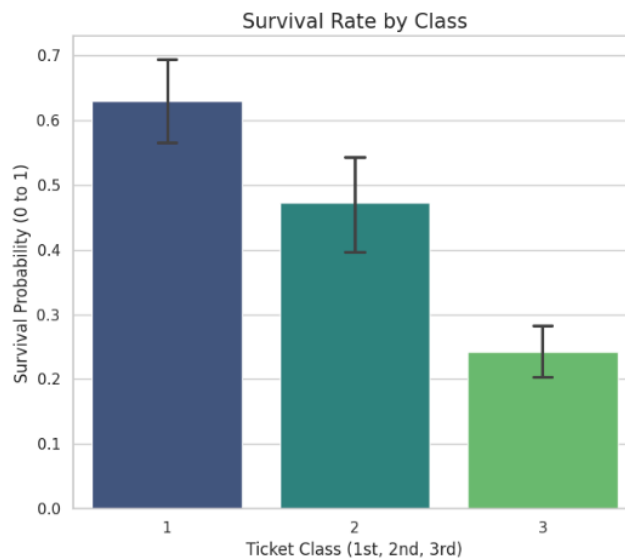


Figure 2.3

Figure 2.1 shows the survival rates of male and female passengers, revealing that most males did not survive, while most females did. **Figure 2.2** presents survival rates across different age groups, highlighting that children were the only group where the number of survivors clearly exceeded the number of deaths, while in all other age groups deaths were much higher. Finally, **Figure 2.3** illustrates survival by passenger class, showing that passengers in 1st and 2nd class were far more likely to survive compared to those in 3rd class, meaning social status plays a big role in survival.

IV. Multivariate Analysis

This section shows how multiple factors combined to affect passenger survival. Visualisations that compare survival rates by class and gender, display survival trends across classes for males and females, and examine how family size influenced survival by grouping passengers into those traveling alone, in small families, or in large families.

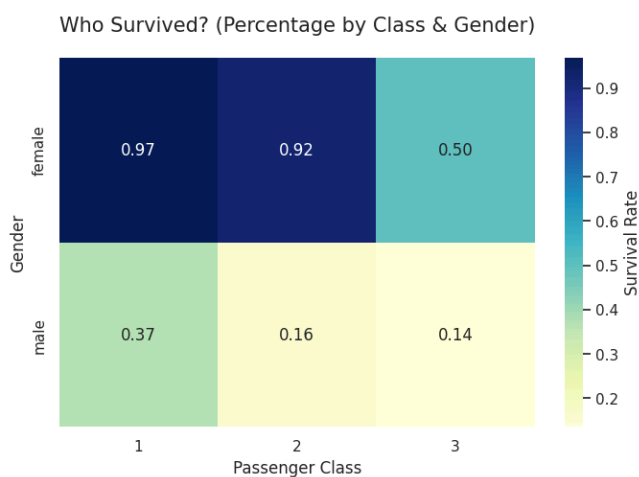


Figure 3.1

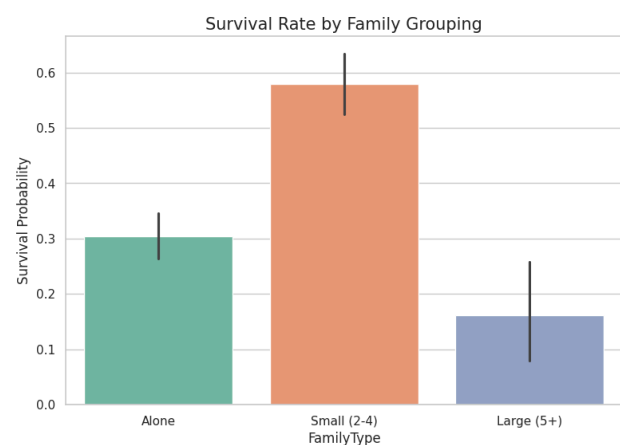


Figure 3.2

The **(Figure 3.1)** heatmap shows survival percentages based on gender and class. It reveals that females in 1st and 2nd class had the highest survival rates. Although females in 3rd class had lower survival, their chances were still much higher than those of males across all classes, highlighting that women were generally prioritized during the sinking. **Figure 3.2** shows a graph of survival rates by family size shows that passengers in small families of 2–4 members had the highest survival, followed by those traveling alone, while large families of 5 or more had the lowest survival. The lower survival rate for large families does not mean they were ignored. It likely reflects that many large families included mostly men, while children and women especially in higher classes were prioritized for lifeboats. Similarly, being alone did not guarantee survival, but many passengers traveling alone were children or women, who were given priority.

V. Observations

This section shows the personal observations based on the patterns on the Dataset:

- **Profile of a Survivor** - The passengers most likely to survive were generally females traveling in 1st or 2nd class, often accompanied by a small family of two to four members. Children, regardless of gender, also had higher survival chances, which reflects the “women and children first” social standard during the evacuation. Higher-class passengers had better survival outcomes overall, likely due to easier proximity to safety points because of the class status on the ship. Traveling in a small family appears to have been an advantage as these groups were neither too large to slow evacuation and not completely alone, which may have allowed them to benefit from prioritization.
- **Profile of a Non Survivor** - Passengers least likely to survive were dominant adult males in 3rd class, often traveling alone or as part of a large family of five or more. Survival rates for men were consistently lower across all classes, highlighting that gender played a major role in prioritization. Large families faced additional challenges. Even if some members were women or children, the majority of men in these groups could have contributed to lower overall survival for the family. Being alone did not guarantee survival either, as most solo travelers could be possibly male.

VI. Additional Insights

The dataset would be more valuable if it included details for all passengers on the Titanic, which would provide a more balanced view and allow verification of whether the observed patterns hold for the complete population. Additionally, adding or combining other features could improve the analysis, as they might reveal further patterns into factors affecting passenger survival.